

Leggett-Garg Inequality-based Quantum Computers Quality Benchmark

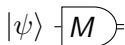
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IQM Awards 2026 Talk
6 VIII 2026, Online

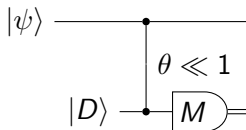
Projective vs weak measurement

Projective (strong)



- Direct
- Accurate (in theory)
- Disturbing (decoherence)

Weak



- Indirect (weak coupling)
- Inaccurate (deliberately)
- small decoherence

No free lunch: information gain $> \theta$, decoherence $> \theta^2$.

Application: Leggett-Garg inequality violation

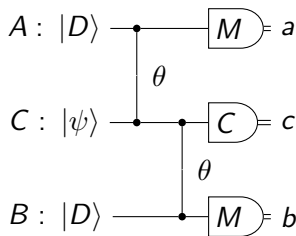
$$\langle a \rangle + \langle b \rangle - \langle ab \rangle \leq 1, |a| \leq 1, |b| \leq 1$$

The catch: strength calibration θ : $a_{raw} = \theta a_{physical}$

- Strength-agnostic form
- postselection $c = 0, 1$,
- $A \equiv B$ (identical, independent)

$$Q = \frac{\langle a + b|c \rangle^2}{4\langle ab|c \rangle} \leq 1$$

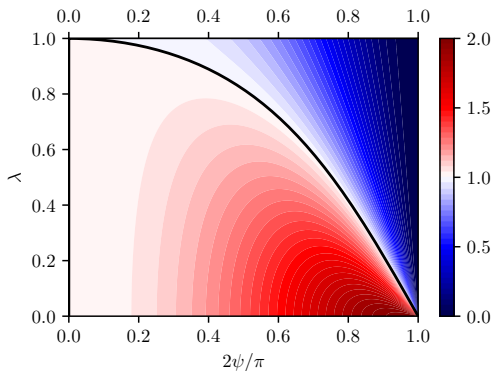
$$\langle ab|c \rangle = \langle abc \rangle / p(c = 1)$$



A test with strength dependence on $\lambda = \sin \theta$

$$A = B = Z = |0\rangle\langle 0| - |1\rangle\langle 1|, \quad C = |\psi_{-}\rangle\langle\psi_{-}|, \quad |\psi\rangle = |\psi_{+}\rangle$$

$$|\psi_{\pm}\rangle = \cos \frac{\psi}{2} |1\rangle \pm \sin \frac{\psi}{2} |0\rangle$$

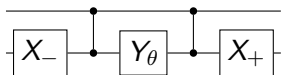


Weak coupling implementation

- two qubit gate $U \sim I$, basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$
- IBM/IonQ – native

$$RZZ(\theta) = \text{diag}(1, e^{i\theta}, e^{i\theta}, 1)$$

- IQM – two CZ sandwiching Y_θ

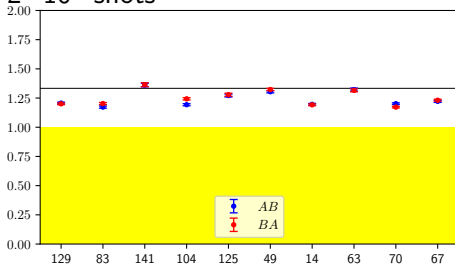


Notation: $X = |0\rangle\langle 1| + |1\rangle\langle 0|$, $Y = i|1\rangle\langle 0| - i|0\rangle\langle 1|$, $V_\theta = \exp(-i\theta V)$,
 $V_\pm = V_{\pm\pi/2}$

Results on IBM/IonQ/IQM $\theta = 0.1$

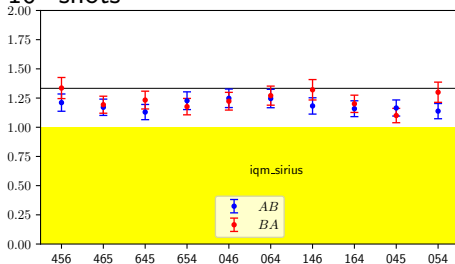
IBM kingston (C qubit specified)

$2 \cdot 10^7$ shots



IQM Sirius (CAB qubits specified)

10^6 shots



IonQ Forte 1: 1.272 ± 0.035 ($A \rightarrow B$), 1.289 ± 0.035 ($B \rightarrow A$)

Impact and roadmap

Challenges

- more access needed to lower error $\sigma \sim 1/\theta^2 \sqrt{N}$,
- pulse control $U \rightarrow U^\theta$, e.g. $CZ \rightarrow \text{diag}(1, 1, 1, e^{\pm i\theta})$, continuous families of entangling gates
- avoid postselection (full noninvasiveness)
- Star topology $\rightarrow$ scale up the test, more detectors

Benefits

- benchmarking partially entangling gates and pulse control
- proposed benchmarking score/map (for the control range)

$$Q - 1, (Q - 1)/\sigma$$

- minimal disturbance: $\theta = 1/10$ disturbance $1/400$.
- robustness to noise – no error correction needed

Thank you for your attention!

https://github.com/Tomev/IQM_Research_Award_2026

Last but not least!

- our long-standing collaborators/supporters: Wolfgang Belzig, Jakub Tworzydło, and Josep Batle
- to read more <https://arxiv.org/abs/2603.22020>
- IBM Quantum, IonQ, IQM – access provided
- Poznań Supercomputing and Networking Center – Polish access hub
- funding "Excellence initiative – research university" for the AGH University in Krakow as well as the ARTIQ project:
UMO-2021/01/2/ST6/00004 and ARTIQ/0004/2021.